```
/* Name-Nirmayee kadam
   /PRN-25070521134/
   SECTION-C-I
   Value Added Problem-2 */
#include <stdio.h>
#include <string.h>

#define MAX 1000000

char stack[MAX];
int top = -1;

// Push function
void push(char c) {
    stack[++top] = c;
}

// Pop function
char pop() {
    if (top == -1)
        return '\0';
    return stack[top--];
}

// Check if matching pair
int isMatchingPair(char open, char close) {
    if (open == '(' && close == ')') return 1;
    if (open == '{' && close == '}') return 1;
```

```
    if (open == '[' && close == ']') return 1;
    return 0;
}

int isBalanced(char *s) {
    for (int i = 0; s[i] != '\0'; i++) {

        // If opening bracket, push
        if (s[i] == '(' || s[i] == '{' || s[i] == '[') {
            push(s[i]);
        }
        // If closing bracket
        else if (s[i] == ')' || s[i] == '}' || s[i] == ']') {

            if (top == -1)  // No matching opening
                return 0;

            char topElement = pop();

            if (!isMatchingPair(topElement, s[i]))
                return 0;
        }
    }

    // If stack is empty → balanced
    return (top == -1);
}
```

```c
int main() {
    printf("Name :Nirmayee kadam \n");
    printf("PRN :25070521134\n");
    printf("Section :C-1\n");
    printf("Value Added Problem : 2\n");
    printf("--------------------------\n");
    char s[MAX];

    printf("Enter expression: ");
    scanf("%s", s);

    if (isBalanced(s))
        printf("true\n");
    else
        printf("false\n");

    return 0;
}
```